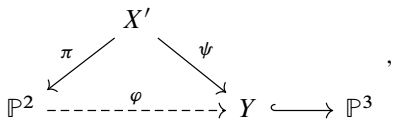
# Chapter 5, Section 4

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**1.** The linear system of conics in $\mathbb{P}^2$ with two assigned base points $P_1$ and $P_2$ (4.1) determines a morphism $\psi$ of $X'$ (which is $\mathbb{P}^2$ with $P_1$ and $P_2$ blown up) to a nonsingular quadric surface $Y$ in $\mathbb{P}^3$, and furthermore $X'$ via $\psi$ is isomorphic to $Y$ with one point blown up.

*Proof.* We have the commutative diagram




where $\varphi$ is defined by the linear system $\mathfrak{d}$ of conics on $\mathbb{P}^2$ with assigned base-points $P_1$ and $P_2$, $Y$ is the image of $\varphi$, and $X'$ is the blow-up of $\mathbb{P}^2$ at $P_1$ and $P_2$. We may take the two points to be $P_1 = (1:0:0)$ and $P_2 = (0:1:0)$. Then the vector space $V \subseteq \mathrm{H}^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(2))$ corresponding to $\mathfrak{d}$ is spanned by

$$y_0 = x_1x_2, \quad y_1 = x_0x_2, \quad y_2 = x_0x_1, \quad y_3 = x_2^2,$$

where $y_i$ are the homogenous coordinates of $\mathbb{P}^3$. In particular, $\varphi$ is defined as

$$\varphi(x_0 : x_1 : x_2) = (y_0 : y_1 : y_2 : y_3),$$

and the points in the image of $\psi$ satisfies the equation

$$y_0y_1 - y_2y_3 = 0,$$

which corresponds to a nonsingular quadric surface $Y \subset \mathbb{P}^3$. It is a birational map since if we restrict to the affine patch defined by $x_2 \neq 0$, then we can identify $\varphi$ with a quadratic transformation (4.2.3).

For each $i = 1, 2$, let $E_i$ be the strict transform of $P_i$, let $M$ be the line through $P_1$ and $P_2$, which is defined by $x_2 = 0$ in $\mathbb{P}^2$, and let $\tilde{M}$ be the strict transform of $M$ in $X'$. By (3.6), we have

$$\tilde{M} \sim \pi^*M - E_1 - E_2,$$

so by (3.2), $\tilde{M}$ has self intersection

$$\tilde{M}^2 = (M)^2 + (E_1)^2 + (E_2)^2 = -1.$$

Let $\lambda$ be a local coordinate on $M$ such that $M$ consists of the points $(\lambda : 1 : 0)$. The image of $M$ under $\varphi$ for $\lambda \neq 0$ is computed to be

$$\varphi(\lambda : 1 : 0) = (0 : 0 : \lambda : 0),$$

so denote by $Q$ the point $\varphi(M - \{P_1, P_2\}) = (0 : 0 : 1 : 0)$. We claim $X'$ can be identified with the blowing-up of $Y$ at $Q$. Indeed, $\varphi$ induces an isomorphism

$$\mathbb{P}^2 - M \cong Y - \{Q\} \cup \psi(E_1) \cup \psi(E_2),$$

and $\pi$ induces an isomorphism

$$X' - E_1 \cup E_2 \cong \mathbb{P}^2 - \{P_1, P_2\},$$

so $\psi$ induces an isomorphism

$$X' - \tilde{M} \cup E_1 \cup E_2 \cong Y - \{Q\} \cup \psi(E_1) \cup \psi(E_2),$$

and since $\psi(E_i)$ are lines in $\mathbb{P}^3$ which intersect at $Q$, we conclude that $\psi$ induces an isomorphism

$$X' - \tilde{M} \cong Y - \{Q\}.$$

Hence by (5.7), $X'$ can be identified with $Y$ blown-up at $Q$. (See [Manin, §18] for more details.) □

**2.** Let $\varphi$ be the quadratic transformation of (4.2.3), centered at $P_1, P_2, P_3$. If $C$ is an irreducible curve of degree $d$ in $\mathbb{P}^2$, with points of multiplicity $r_1, r_2, r_3$ at $P_1, P_2, P_3$, then the strict transform $C'$ of $C$ by $\varphi$ has degree $d' = 2d - r_1 - r_2 - r_3$, and has points of multiplicity $d - r_2 - r_3$ at $Q_1$, $d - r_1 - r_3$ at $Q_2$, and $d - r_1 - r_2$ at $Q_3$. The curve may have arbitrary singularities.

*Proof.* We have a commutative diagram

$$\begin{array}{ccc} & X & \\ {\scriptstyle\pi}\swarrow & & \searrow{\scriptstyle\psi} \\ \mathbb{P}^2 & \overset{\varphi}{\dashrightarrow} & \mathbb{P}^2 \end{array}.$$

Let $H$ be a hyperplane section of $\mathbb{P}^2$, so that $\varphi$ is given by the linear system $\mathfrak{d} = |2H - P_1 - P_2 - P_3|$, and let $E_1, E_2, E_3$ be the exceptional divisors of $X$ via the blowing-up of $\mathbb{P}^2$ at $P_1, P_2, P_3$, respectively. By (3.6), the strict transform $\tilde{C}$ of $C$ by $\pi$ is linearly equivalent to

$$\tilde{C} = \pi^* C - r_1 E_1 - r_2 E_2 - r_3 E_3.$$

Since $\psi$ is given by the linear system $\mathfrak{d}' = |D|$ with $D = 2\pi^* H - E_1 - E_2 - E_3$, $\varphi$ is a birational map such that $C' = \varphi(C) = \psi(\tilde{C})$, and $C \sim dH$ in $\mathbb{P}^2$, the degree of $C'$ in $\mathbb{P}^2$ can be computed using (Ex. 1.2) and (3.2)

$$\begin{aligned} \tilde{C}.D &= (d\pi^* H - r_1 E_1 - r_2 E_2 - r_3 E_3).(2\pi^* H - E_1 - E_2 - E_3) \\ &= 2d(H^2) + r_1(E_1)^2 + r_2(E_2)^2 + r_3(E_3)^2 \\ &= 2d - r_1 - r_2 - r_3. \end{aligned}$$

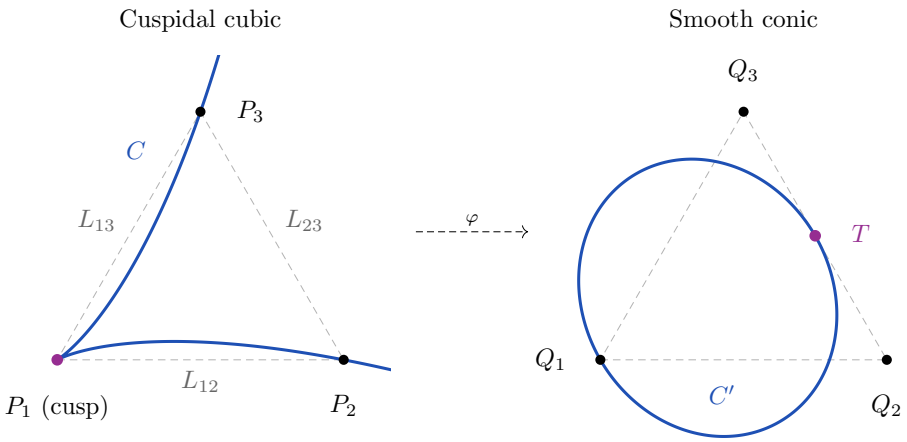
For each $i \neq j$, let $L_{ij}$ be the line which goes through $P_i$ and $P_j$. By (3.6), for each $k = 1, 2, 3$, the multiplicity of $C'$ at $Q_k$ is the intersection of $\tilde{C}$ with the strict transform $\tilde{L}_{ij}$ of $L_{ij}$ with $i, j \neq k$ via $\pi$ (4.2.3), which by (3.6) is linearly equivalent to

$$\tilde{L}_{ij} = \pi^* L_{ij} - E_i - E_j.$$

Again, we have $L_{ij} \sim H$ in $\mathbb{P}^2$, so

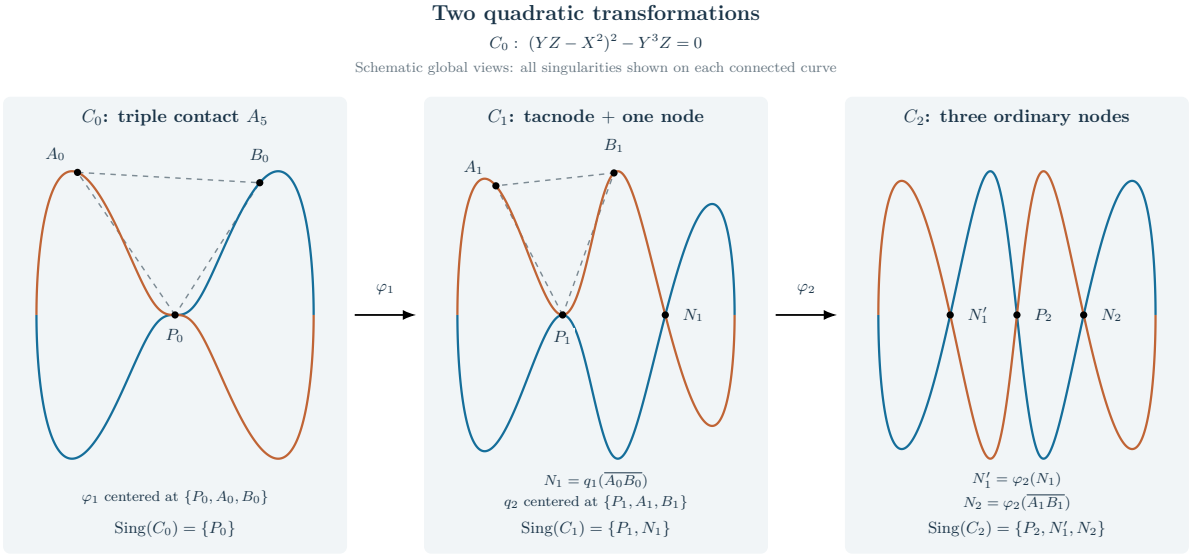
$$\begin{aligned} \tilde{C}.\tilde{L}_{ij} &= (d\pi^* H - r_1 E_1 - r_2 E_2 - r_3 E_3).(\pi^* H - E_i - E_j) \\ &= d(H)^2 + r_i(E_i)^2 + r_j(E_j)^2 \\ &= d - r_i - r_j. \end{aligned}$$

Hence, $C'$ has points of multiplicity $d - r_i - r_j$ at $Q_k$. □




**3.** Let $C$ be an irreducible curve in $\mathbb{P}^2$. Then there exists a finite sequence of quadratic transformations, centered at suitable triples of points, so that the strict transform $C'$ of $C$ has only *ordinary* singularities, i.e., multiple points with all distinct tangent directions.

*Proof.* If $C$ is a conic in $\mathbb{P}^2$, then it is nonsingular, so assume $C$ has degree $\geq 3$, and let $P_1$ be a nonordinary singular point of $C$. Our goal is to find points $P_2, P_3 \in \mathbb{P}^2$ such that the quadratic transformation centered at $P_i$ improves the singularity at $P_1$ and only introduces ordinary singularities on the strict transform of $C$. To that end, for each $i \neq j$, let $L_{ij}$ be the line through $P_i$ and $P_j$. The restriction of the quadratic transformation $\varphi : \mathbb{P}^2 \to \mathbb{P}^2$ centered at $P_i$ to $\mathbb{P}^2 - \bigcup L_{ij}$ is an isomorphism onto its image, and each $L_{ij}$ is sent to a point $Q_k$ with $k \neq i, j$ such that if $M$ is a curve meeting $L_{ij}$ transversally away from the centers, then the strict transform of $M$ has ordinary singularities at $Q_k$. Therefore, we require that away from $P_1$, the lines $L_{ij}$ meets $C$ transversally, and $L_{12}, L_{13}$ are distinct from the tangent directions of $C$ at $P$. There are only finitely many tangent directions of $C$ at $P_1$, and since $C$ is not strange (IV, 3.8) (IV, 3.9), we can find many lines $L_{12}$ and $L_{13}$ through $P_1$ with tangent vectors distinct from that of $C$ at $P_1$ which meets $C$ transversally. Lastly, pick $L_{23}$ which avoids $P_1$, not parallel to $L_{12}, L_{13}$, meets $C$ transversally, and the points $P_2, P_3$ of intersection of $L_{23}$ with $L_{12}, L_{13}$, respectively, are not on $C$, which again exists because $C$ is not strange. The strict transform of $C$ under the quadratic transformation centered at $P_i$ improves the singularity at $P_1$ (3.7) and only introduces ordinary singularities at $Q_k$. By (3.9), we can repeat this process for each nonordinary singularity until the strict transform only has ordinary singularities introduced by the quadratic transformations. □




**4.** (a) Use (4.5) to prove the following lemma on cubics: If $C$ is an irreducible plane cubic curve, if $L$ is a line meeting $C$ in points $P, Q, R$, and $L'$ is a line meeting $C$ in points $P', Q', R'$, let $P''$ be the third intersection of the line $PP'$ with $C$, and define $Q''$, $R''$ similarly. Then $P'', Q'', R''$ are collinear.

(b) Let $P_0$ be an inflection point of $C$, and *define* the group operation on the set of regular points of $C$ by the geometric recipe "let the line $PQ$ meet $C$ at $R$, and let $P_0R$ meet $C$ at $T$, then $P + Q = T$" as in (II, 6.10.2) and (II, 6.11.4). Use (a) to show that this operation is associative.

*Proof.*

(a) Let $L_P$ be the line $PP'$, and similarly for $L_Q, L_R$, and let $M$ be the line going through $P''$ and $Q''$. The divisors $C$ and $L_P + L_Q + L_R$ belong to the linear system of conics in $\mathbb{P}^2$ with assigned base points $P, P', P'', Q, Q'', Q''', R, R'$, and they intersect at $R''$, so by (4.5) $R''$ is the unique point in $\mathbb{P}^2$ such that every cubic curve in $\mathbb{P}^2$ going through $P, P', P'', Q, Q'', Q''', R, R'$ also goes through $R''$. In particular, $L + L' + M$ belongs to this linear system, so it must go through $R''$. Therefore, $M$ must go through $R''$.

(b) Let $P_1, P_2, P_3$ be regular points of $C$, let $L_{12}$ be the line $P_1P_2$, let $R_{12}$ be the third point $L_{12}$ meets $C$, let $L'_{12}$ be the line $P_0R_{12}$, let $Q_{12} = P_1 + P_2$, and similarly for $L_{23}, L'_{23}$, and $R_{23}, Q_{23}$. Let $M_{12,3}$ be the line $Q_{12}P_3$, let $S_{12,3}$ be the third point $M_{12,3}$ meets $C$, let $M'_{12,3}$ be the line $P_0S_{12,3}$, let $T_{12,3} = Q_{12} + P_3$, and similarly for $M_{1,23}, M'_{1,23}$, and $S_{1,23}, T_{1,23}$. We want to show $T_{12,3} = T_{1,23}$. Applying (a) to

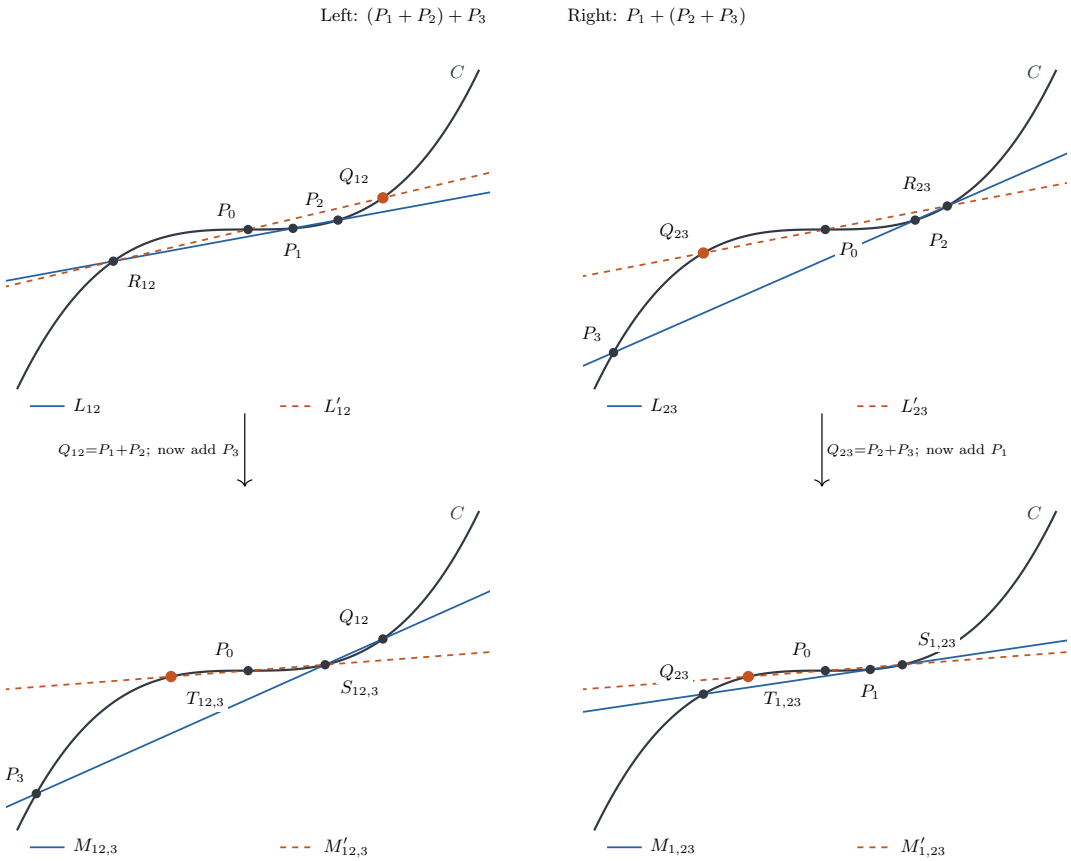
$$L = L_{12}, \quad L' = L'_{23},$$

and

$$\begin{aligned} PP' &= M_{1,23}, & P'' &= S_{1,23}, \\ QQ' &= L_{23}, & Q'' &= P_3, \\ RR' &= L'_{12}, & R'' &= Q_{12}, \end{aligned}$$

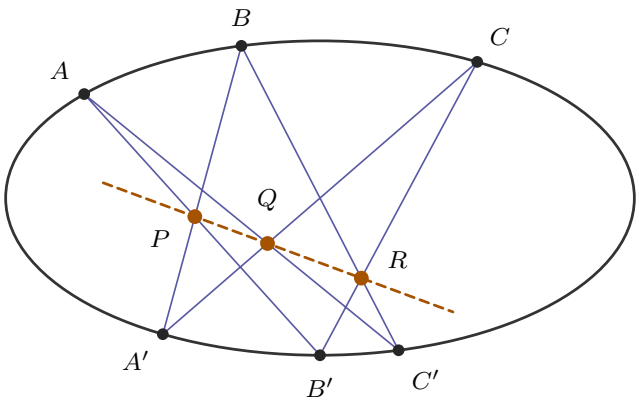
shows $S_{1,23}$, $P_3$, and $Q_{12}$ are collinear. Therefore, $S_{1,23} = S_{12,3}$, and so $T_{1,23} = T_{12,3}$.

□




$$T_{12,3} = (P_1 + P_2) + P_3 = P_1 + (P_2 + P_3) = T_{1,23}.$$

**5.** Prove Pascal's theorem: if $A, B, C, A', B', C'$ are any six points on a conic, then the points $P = AB'.A'B$, $Q = AC'.A'C$, and $R = BC'.B'C$ are collinear.




*Proof.* Denote the conic by $C$, and let $L$ be the line $PQ$. By (4.5), any cubic curve through the points $A, B, C, A', B', C', P, Q$ passes through a uniquely determined point, which we claim to be $R$. Indeed,

the divisors $AB' + A'C + BC'$ and $AC' + A'B + B'C$ intersect at $R$, so $C + L$ passes through $R$. Hence, $L$ passes through $R$. □

**12.** Use (4.11) to show that if $D$ is any ample divisor on the cubic surface $X$, then $\mathrm{H}^1(X, \mathcal{O}_X(-D)) = 0$. This is Kodaira's vanishing theorem for the cubic surface (III, 7.15).

*Proof.* By (4.11), $D$ is ample if and only if $D$ is very ample, so by Bertini's theorem, we can assume $D$ is irreducible and nonsingular as a closed subvariety of $X$. In particular, $\mathcal{O}_X(-D)$ has no global sections, and we have a short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0 \ ,$$

where $\mathcal{O}_D$ is the structure sheaf of $D$ as a variety. Since $X$ is a hypersurface in $\mathbb{P}^3$,

$$h^1(X, \mathcal{O}_X) = h^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) = 0,$$

so the long exact sequence in cohomology gives

$$0 \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X) \longrightarrow \mathrm{H}^0(D, \mathcal{O}_D) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X(-D)) \longrightarrow 0 \ ,$$

and since $D$ is irreducible,

$$h^0(X, \mathcal{O}_X) = h^0(D, \mathcal{O}_D) = 1.$$

Hence by exactness, $\mathrm{H}^1(X, \mathcal{O}_X(-D)) = 0$. □

**13.** Let $X$ be the Del Pezzo surface of degree 4 in $\mathbb{P}^4$ obtained by blowing up 5 points of $\mathbb{P}^2$ (4.7).

(a) Show that $X$ contains 16 lines.

(b) Show that $X$ is a complete intersection of two quadric hypersurfaces in $\mathbb{P}^4$ (the converse follows from (4.7.1)).

*Proof.*

(a) *Claim.* The 16 lines of $X$ are

(i) the exceptional curves $E_i \sim e_i$, $i = 1, \ldots, 5$ (five of these),

(ii) the strict transform $F_{ij} \sim \ell - e_i - e_j$ of the line in $\mathbb{P}^2$ containing $P_i$ and $P_j$, $1 \le i < j \le 6$ (ten of these), and

(iii) the strict transform $G \sim 2\ell - \sum_{i=1}^5 e_i$ of the conic in $\mathbb{P}^2$ containg the five $P_i$ (one of these).

Adjusting accordingly, the statements of (4.8) hold for $X$ as well. If $C \sim a\ell - \sum_{i=1}^5 b_i e_i$ is a curve in $X$, then by (4.8)

$$p_a(C) = \frac{1}{2}(C^2 - \deg C) + 1,$$

so $C$ is a line in $X$ if and only if $\deg C = 1$ and $p_a(C) = 0$ if and only if $C$ is an irreducible curve on $X$ with $C^2 < 0$.

Now, suppose $C$ is any irreducible curve on $X$ with $\deg C = 1$ and $C^2 = -1$. We want to show $C$ is one of the 16 lines listed. Assuming $C$ is not one of the $E_i$, we must have $a > 0$ and $b_i \ge 0$ by (4.8.1). Furthermore,

$$\deg C = 3a - \sum_{i=1}^5 b_i = 1$$

$$C^2 = a^2 - \sum_{i=1}^5 b_i^2 = -1.$$

The proof of (4.9) follows verbatim, so Schwarz's inequality implies

$$(\sum_{i=1}^5 b_i)^2 \le 5(\sum_{i=1}^5 b_i^2).$$

Substituting $\sum_{i=1}^5 b_i = 3a - 1$ and $\sum_{i=1}^5 b_i^2 = a^2 + 1$ from above, we obtain

$$(2a + 1)(a - 2) \leq 0,$$

which implies $a \leq 2$. Therefore, $a = 1$ or 2. If $a = 1$, then

$$\sum_{i=1}^5 b_i = 2, \quad \sum_{i=1}^5 b_i = 2,$$

which gives $F_{ij}$ for some $i \neq j$. If $a = 2$, then

$$\sum_{i=1}^5 b_i = 5, \quad \sum_{i=1}^5 b_i^2 = 5,$$

and the only possible values are $b_i = 1$ for every $i = 1, \ldots, 5$, which gives $G$.

(b) Let $\mathcal{I}_X$ be the ideal sheaf of $X$, and consider the short exact sequence

$$0 \longrightarrow \mathcal{I}_X \longrightarrow \mathcal{O}_{\mathbb{P}^4} \longrightarrow \mathcal{O}_X \longrightarrow 0 \ .$$

Twisting by $\mathcal{O}_{\mathbb{P}^4}(2)$ and taking global sections give

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^4, \mathcal{I}_X(2)) \longrightarrow \mathrm{H}^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(2)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(2)) \ .$$

The vector space in the middle has dimension $\binom{4+2}{2} = 15$. On the other hand, $\mathcal{O}_X(2) = \mathcal{O}_X(-2K)$, where $K$ is a canonical divisor of $X$, so Riemann-Roch gives

$$h^0(X, \mathcal{O}_X(-2K)) - h^1(X, \mathcal{O}_X(-2K)) + h^2(X, \mathcal{O}_X(-2K)) = 13,$$

and by Serre duality,

$$h^2(X, \mathcal{O}_X(-2K)) = h^0(X, \mathcal{O}_X(3K)) = h^0(X, \mathcal{O}_X(-3)) = 0,$$

so it remains to compute $h^1(X, \mathcal{O}_X(-2K))$. The Leray spectral sequence

$$\mathrm{H}^i(\mathbb{P}^2, \mathrm{R}^j \pi_* \mathcal{O}_X(-2K)) \implies \mathrm{H}^{i+j}(X, \mathcal{O}_X(-2K))$$

degnerates, so we have $h^1(X, \mathcal{O}_X(-2K)) = 0$. Combining everything gives $h^0(X, \mathcal{O}_X(2)) = 13$, which means

$$\dim \mathrm{H}^0(\mathbb{P}^4, \mathcal{I}_X(2)) \geq 2.$$

*The author will fill in the details of the Leray spectral sequence calculation in the future, after learning what they are.* An element of that space is a form of degree 2, whose zero-set will be a quadric hypersurface containing $X$, and there exists at least two distinct irreducible quadric hypersurfaces containing $X$. Hence by reasons of degree, $X$ is a complete intersection of two quadric hypersurfaces.

□

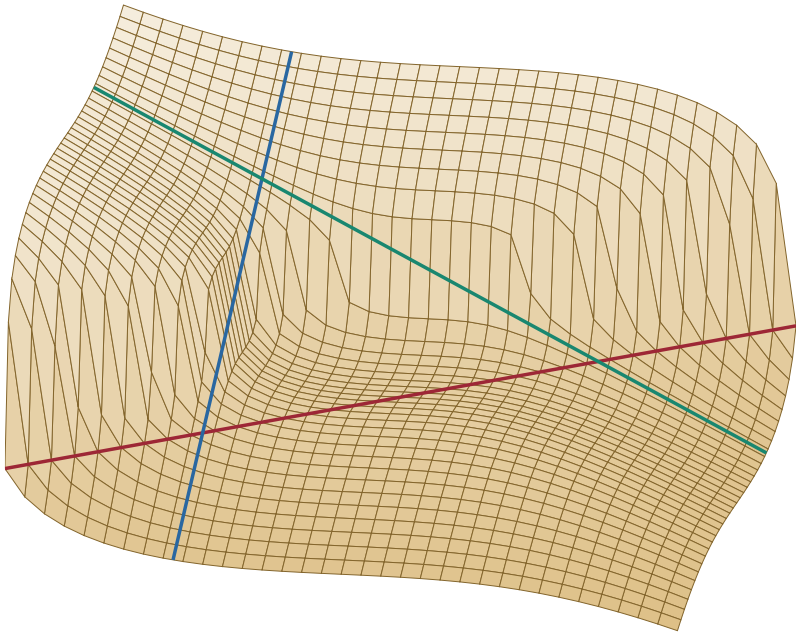
**14.** Using the method of (4.13.1), verify that there are nonsingular curves in $\mathbb{P}^3$ with $d = 8$, $g = 6, 7$; $d = 9$, $g = 7, 8, 9$; $d = 10$, $g = 8, 9, 10, 11$. Combining with (IV, §6), this completes the determination of all possible $g$ for curves of degree $d \geq 10$ in $\mathbb{P}^3$.

*Proof.*

| $d$ | $g$ | $a$ | $(b_1, \ldots, b_6)$ |
|---|---|---|---|
| 8 | 6 | 7 | (3, 3, 2, 2, 2, 1) |
| 8 | 7 | 7 | (3, 2, 2, 2, 2, 2) |
| 9 | 7 | 7 | (3, 3, 2, 2, 1, 1) |
| 9 | 8 | 7 | (3, 2, 2, 2, 2, 1) |
| 9 | 9 | 7 | (2, 2, 2, 2, 2, 2) |
| 10 | 8 | 7 | (3, 3, 2, 1, 1, 1) |
| 10 | 9 | 7 | (3, 2, 2, 2, 1, 1) |
| 10 | 10 | 7 | (2, 2, 2, 2, 2, 1) |
| 10 | 11 | 8 | (3, 3, 2, 2, 2, 2) |

□

**16.** For the *Fermat cubic surface.* $x^3 + y^3 + z^3 + w^3 = 0$, find the equations of the 27 lines explicitly, and verify their incidence relations. What is the group of automorphisms of this surface?



$$x^3 + y^3 + z^3 + 1 = 0 \qquad (w = 1)$$

*Proof.* Let $\omega^3 = 1$ be a primitive third root of unity in $k$. The line defined by the parametric equation

$$t \mapsto (\omega^i t, -\omega^j t, -\omega^k), \quad i, j, k = 0, 1, 2,$$

is contained in $X$. There are 27 possible combinations of $(i, j, k)$, and $(i, j, k), (i + 1, j + 1, k + 1)$ define the same lines in $\mathbb{P}^3$, which gives nine of the lines. By symmetry of the defining equation, we obtain the rest.

The automorphisms of $X$ are the following. Any permutation of the coordinates preserve the defining equation, which is an action of the symmetric group $\Sigma_4$. Also, multiplication of any of the coordinates by any power of $\omega$ also preserve the defining equation, which is an action of $(\mathbb{Z}/3)^{\oplus 4}/\mathbb{Z}(1, 1, 1, 1) \cong (\mathbb{Z}/3)^{\oplus 3}$. Hence, $\operatorname{Aut} X \cong \Sigma_4 \rtimes (\mathbb{Z}/3)^{\oplus 3}$, which has size 648. □